

CashBook: Empowering small businesses in India and beyond with fintech solutions powered by Firebase

Tell us your challenge. We're here to help.

Contact us (/contact)



About CashBook

CashBook built a
empowering finte
small businesses
its customer bas
800,000 within si
launched secure

CashBook (<https://cashbook.in/>) is an Indian fintech startup committed to empowering small businesses with the "simplicity of technology." Its innovative mobile app enables real-time, collaborative record keeping of business transactions, opening the door to financing opportunities through big data analytics. Since launching in 2018, CashBook has grown to more than 2,000 Indian cities, with 800,000 active monthly users, for \$1.5 billion monthly recorded transactions. In 2021, CashBook raised \$2.3 million in a seed funding round.

solutions with financial data, and forged global expansion with Google Cloud

Industries: Financial Services & Insurance

Location: India

Firebase (<https://firebase.google.com/>), Cloud

Google Cloud results

- Enables rapid expansion with lean development team by halving backend provisioning needs
- Empowers customers with real-time data insight through two-fold latency boost
- Augments personalized solutions by reducing time

Double
customer
in 6 months
zero cost



About Niveus Solutions

Niveus Solutions is a Google Cloud Partner on a people-centric mission to enable Indian innovators from healthcare to finance and entertainment to optimize their digital transformation. In 2020, it was named Asia-Pacific Google Cloud Partner of the Year.

About Niveus

Google Cloud (<https://cloud.google.com/>)

Solutions

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Firebase (<https://firebase.google.com/>)

Indian innovators from healthcare to

Cloud Functions

(<https://cloud.google.com/functions/>)

Cloud Firestore

(<https://cloud.google.com/firestore/>)

required to
identify
power users
from 1
month to 1
week

- Supports
global
growth
through
linkage to
worldwide
data center
network
- Enables
collaborative
solution
building
thanks to
advanced
security
protocols

As university students in India, Vivek Kumar and Ashutosh Pathak ran a laundromat business to get a headstart in their dreams of entrepreneurial success. The experience gave them first-hand knowledge of the challenges small business owners face in keeping track of finances while juggling responsibilities.

Speaking with fellow business owners, they discovered that the problems they faced were widespread in India, and that inefficient pen-and-

paper bookkeeping methods had a knock-on effect of preventing business people from accessing financing due to a lack of reliable digitized records. The experience inspired them to found CashBook (<https://cashbook.in/>), a mobile app that enables small business owners to easily track transactions from income and expenses to loan payments, with a net balance at the end of each day.

To drive their mission of business empowerment, Kumar and Pathak needed a powerful cloud solution to enable unlimited scale with a lean development team, as well as built-in security features for handling sensitive financial data. After surveying major cloud providers, they chose to build their mobile app on Firebase (<https://firebase.google.com/>), a Google Cloud (<https://cloud.google.com/>) managed platform specifically designed to enable developers to build web and mobile applications with ease.

"The simplicity and power of Firebase are determining factors in our decision to choose Google Cloud," says Ashutosh Pathak, co-founder and CTO, CashBook. "My experience with a rival cloud provider often left me confused as to what service I should use, or what the dashboards were saying. Everything on Google Cloud is so easy to understand, with intuitive dashboards that we can just get down to the business of creating the app."

Unlocking limitless growth potential with Firebase managed services and creative partnership

CashBook took off across India and beyond, thanks to the app's own intuitive user interface, and a unique feature that allows secure, team-driven bookkeeping. Since launching in 2018, the startup has enjoyed a strong pace of growth to reach 800,000 monthly active users, some as far away as Africa. Pathak says it's a journey made possible by the unlimited auto-scaling and near-zero latencies of Firebase, as well as working with Niveus Solutions (<https://niveussolutions.com/>).

As CashBook grew on Firebase and started to add more complex features, the startup began to deploy Cloud Functions for Firebase

(<https://firebase.google.com/docs/functions>) as a serverless framework to automatically run backend code in response to events from app users. This provided a crucial advantage in freeing up the engineering team from the chore of designing backend protocols, so it could concentrate on creative app development.

"By building our mobile app on Firebase and Cloud Functions, we can focus on the core business logic, while maintaining a lean team of engineers," says Pathak. "With powerful managed services, I never have to lose sleep at night wondering if the app is up. Firebase and Cloud Functions are constantly scaling up and managing the back end to meet demand."

A key driver of CashBook is the ability to crunch millions of data points for business intelligence that enables the startup to better serve customers with tailor-made services and app features. For this,

CashBook relies on Cloud Pub/Sub
(<https://cloud.google.com/pubsub>) to create powerful
data pipelines running from Firestore
(<https://cloud.google.com/firestore>) (its auto-scaling
NoSQL database) into BigQuery
(<https://cloud.google.com/bigquery>) for big data
predictive analytics.

"By building our
mobile app on
Firebase and
Cloud Functions,
we can focus on
the core
business logic,
while
maintaining a
lean team of
engineers. With
powerful
managed
services, I never
have to lose
sleep at night
wondering if the
app is up."

—Ashutosh Pathak, Co-
Founder and CTO,
CashBook

An integrated ecosystem of Google Cloud managed services including Firebase, Firestore, Cloud Functions for Firebase, Cloud Pub/Sub and BigQuery has yielded significant results. According to Pathak, CashBook requires less than half the backend resources it needed developing apps in an earlier on-premise environment. He says it also gained a two-fold boost in latency even at times of peak demand, a key factor for a startup that needs to provide customers real-time insight into their transaction history. Overall, the seamless autoscaling of Google Cloud has enabled CashBook to double its customer base within six months to 800,000 monthly active users, with no loss of latency and zero platform outages.

Since CashBook seeks to be a global solution, and already has a strong presence in Africa and other developing countries in Southeast Asia, it needs to maintain its latency advantages beyond the Indian market. Firebase satisfies this requirement through its global reach, connecting automatically to the nearest Google Cloud data center anywhere in the world.

"With Firebase available worldwide, we are able to provide customers instant and frictionless transaction insight wherever they may be," says Pathak. "This empowers us with unlimited

possibilities of global expansion and creative app development."

"With Firebase
available
worldwide, we
are able to
provide
customers
instant and
frictionless
transaction
insight wherever
they may be.
This empowers
us with
unlimited
possibilities of
global expansion
and creative app
development."

—Ashutosh Pathak, Co-
Founder and CTO,
CashBook

Enabling collaborative data synergies with world class Google Cloud security

For CashBook customers, a critical consideration is the security of their financial data. They need to know that they are able to collaborate securely on CashBook, knowing their sensitive data is safe from prying eyes. In addition, CashBook's development team requires secure data analytics of hundreds of thousands of customer journeys to continuously develop new app solutions and upgrades.

According to Pathak, Google Cloud's secure-by-design services facilitate the handling of millions of sensitive financial data points every day, for the peace of mind of both customers and the CashBook team. "Our users are very concerned about the security of their data. They pepper us with questions about how we ensure data integrity," says Pathak. "Once we tell them it all lives within the Google Cloud ecosystem, they're immediately reassured that the data is safe."

He adds that the robust security protocols of solutions such as Firestore and BigQuery enable the CashBook team to manage sensitive data with no worry of breaches. "Firestore has elaborate setups regarding security, where we can write our own security tests for extra reassurance," says Pathak. "This gives us the confidence to deploy and develop very fast. The ecosystem is built in such a way that we can ship out great products quickly, while maintaining the optimal level of data security that our customers demand."

"Our users are very concerned about the security of their data. They pepper us with questions about how we ensure data integrity. Once we tell them it all lives within the Google Cloud ecosystem, they're immediately reassured that the data is safe."

—Ashutosh Pathak, Co-
Founder and CTO,
CashBook

**Championing the future of fintech
innovation with BigQuery and
Vertex AI**

In the next stage of its evolution, CashBook plans to offer tailor-made financing and credit services based on predictive analytics of customer data. To drive this mission, the startup is working with Niveus Solution to build advanced data modeling with a combination of BigQuery and Vertex AI. Seeing themselves as "young technical champions," Kumar and Pathak are proactively fostering a culture of creative challenge in the young CashBook team, to spark imaginative solutions brainstorming with Niveus Solutions and Google Cloud.

"We're excited to be able to empower more small businesses, in more remote Indian regions, by championing AI solutions such as Google speech-to-text that make the app easier to use for those who struggle with mobile keyboards," says Pathak.

"Google Cloud solutions such as BigQuery, Vertex AI and Natural Language API promise to be key enablers of our mission to empower small businesses across India and beyond."